

For linear map  $f: \mathbb{R}^c \rightarrow \mathbb{R}^r$  we have the following equivalences  
 $x \mapsto Ax$

①  $f$  surjective  $\Leftrightarrow$  For each  $b \in \mathbb{R}^r$   
 $Ax=b$  has a sol<sup>n</sup>  $\Leftrightarrow$  Columns of  $A$   
span  $\mathbb{R}^r$   $\Leftrightarrow$  Rows of  $A$   
are indep  $\Leftrightarrow$  REF( $A$ ) has  
a pivot in  
each row

②  $f$  is injective  $\Leftrightarrow Ax=0$  has only the  
trivial solution  $x=0$   $\Leftrightarrow$  Columns of  $A$   
are independent  $\Leftrightarrow$  Rows of  $A$   
span  $\mathbb{R}^c$   $\Leftrightarrow$  REF( $A$ ) has  
a pivot in  
each column

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# of pivots in REF( $A$ )  $\leq \min(r, c)$  Therefore  $\left\{ \begin{array}{l} \text{I} \Rightarrow r \leq c \\ \text{II} \Rightarrow r \geq c \end{array} \right\} \Rightarrow \text{Any basis of } \mathbb{R}^n$   
 $= n$

$A = r \times c$  matrix  $\xrightarrow[\text{row op}^n]{\text{elem}}$  gives matrix  $[EA]$  where  $E =$  matrix obtained

Claim by doing the same row op<sup>n</sup> or Identity matrix  $I_{r \times r}$

Pf. Note:  $E_{ij}A = \begin{cases} \text{all rows except } i\text{th row are } 0 \\ j\text{th row of } A \text{ moves to } i\text{th row} \end{cases}$

$\swarrow$   
( $r \times r$  matrix unit  
with 1 is slot  $i, j$   
rest 0)

Cases

①  $R_i + aR_j : A \rightarrow (I + aE_{ij})A \quad \checkmark$

②  $cR_i \quad c \neq 0 : A \rightarrow (I + (c-1)E_{ii})A \quad \checkmark$

③  $R_i \leftrightarrow R_j : A \rightarrow (I - E_{ii} - E_{jj} + E_{ij} + E_{ji})A \quad \checkmark$

Consequence:  $A \xrightarrow{\text{row ops}} (E_N \cdots E_2 E_1 A) \rightarrow$  call this matrix  $B$

Columns of  $A$  and cols of  $B$  satisfy  
the exact same set of linear dependence

relations. i.e.  $\{x | Ax=0\} = \{x | Bx=0\}$  as  $Ax=0 \iff Bx=0$

(applies to subsets of cols too!)

T3P1 For a matrix in RFF  $\left\{ \begin{array}{l} \text{pivot cols form a basis for Col space} \\ \text{pivot rows (i.e. non zero rows) form a basis for row space} \end{array} \right. \left\{ \begin{array}{l} \text{lin indep} \checkmark \\ \text{span} \checkmark \end{array} \right.$

How to find basis of  $\text{Row}(A)$ ,  $\text{Nul}(A)$ ,  $\text{Col}(A)$ ?

Answers

Nonzero (i.e. pivot) rows  
of  $\text{REF}(A)$

Magic vectors  
given by  
row reduction  
procedure.

Cols of  $A$  corresponding to  
pivot cols of  $\text{REF}(A)$

Names

$B_R$

$B_N$

$B_C$

Thm!

$$|B_R| = |B_C| = \# \text{ pivots in } \text{REF}(A) \quad \text{"row rank} = \text{col rank"}$$

$$|B_C| + |B_N| = \# \text{ cols of matrix}$$

"rank-nullity thm"

(Captures uniformity  
of fibers of a  
linear map)

Invariance of dimension for f-dim v-spaces

Spl cases ① Suppose  $\{v\}$  &  $\{w_1, w_2\}$  are bases for  $V$ .

$$\therefore \left. \begin{array}{l} w_1 = c_1 v \\ w_2 = c_2 v \end{array} \right\} \Rightarrow c_2 w_1 - c_1 w_2 = 0 \quad \text{contradicts independence of } \{w_1, w_2\}$$

② If  $\{v_1, v_2\}$  &  $\{w_1, w_2, w_3\}$  are bases

$$\left. \begin{array}{l} w_1 = a_1 v_1 + a_2 v_2 \\ w_2 = b_1 v_1 + b_2 v_2 \\ w_3 = c_1 v_1 + c_2 v_2 \end{array} \right\} \left[ \begin{array}{cc} v_1 & v_2 \end{array} \right] \left[ \begin{array}{ccc} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{array} \right] = \left[ \begin{array}{ccc} w_1 & w_2 & w_3 \end{array} \right]$$

Meaning of equation in quotes?

After that?